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19

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JLVX

 Will be reviewed

8

In Go-back-N protocol, what are the size of the respective sender and receiver buffers required for a window size of N?

- A) 1; 1
- B) $N - 1$; 1
- C) N; N
- D) N; 1 *

☐ A

☐ B

☐ C

☒ D

 Will be reviewed

Assume a sender and a receiver connected via one packet-switch. The sender sends a message of size 24 KBytes using packets of length 1000 Bytes. The bit rates are 2 kbps and 5 kbps for the first and the second link, respectively. Using store and forward, what is the end-to-end delay (in seconds)? You may ignore the propagation delay!

- A) 16.8
- B) 134.4
- C) 124
- D) 16800 *

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

10

Consider sending a 1500-byte IP datagram into a link that has an MTU of 500 bytes. Suppose that IP header is 20 bytes long. How many fragments will be generated?

- A) 3
- B) 4
- C) 5
- D) 2 *

☐ A

☒ B

☐ C

☐ D

 Will be reviewed

11

The following diagram shows a simple network topology with 4 nodes. The links in the diagram are labeled with the cost of each link. The nodes run distance vector routing protocol. The protocol has just started, at node X, what is the cost to node z?

- A) 6
- B) 3
- C) 5
- D) 23 *

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

12

Router R3 received two datagrams, which router to deliver it to if the datagrams has a destination IP address 200.23.19.3 and 200.23.18.33?

- A) R2, R1
- B) R2, R2
- C) R1, R1
- D) R1, R2 *

☐ A

☒ B

☐ C

☐ D

 Will be reviewed

13

Which of the following protocols can be used to get the mappings of IP address and MAC address of other nodes in a different subnet?

- A) ARP
- B) DNS
- C) DHCP
- D) None of the options *

☒ A

☐ B

☐ C

☐ D

 Will be reviewed

14

Consider the transmission between a UDP sender and a UDP receiver. Which of the following will never happen? You may assume that the application riding on UDP doesn't implement any reliability mechanisms.

- A) UDP receiver fails to receive any packet from UDP sender.
- B) UDP receiver receives out-of-order packets from UDP sender.
- C) UDP receiver receives duplicate packets from UDP sender.
- D) UDP receiver receives corrupted packets from UDP sender but fails to detect bit errors. *

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

15

Can HTTP response message contains an empty body? If yes, When can this happen?

- A) No, an HTTP response message can never have an empty body.
- B) Yes, if the requested object has been moved to a new location.
- C) Yes, if the requested object is recently modified.
- D) Yes, if the requested object is very small in size. *

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

16

Which of the following statements is true?

- A) Both the Internet and traditional telephone networks use packet-switching.
- B) Performance metrics such as end-to-end delay and throughput can be guaranteed in the Internet, if TCP is chosen as the transport layer protocol.
- C) In the Internet, packets from the same source always take the same path to reach destination.
- D) The Internet uses packet switching and traditional telephone networks uses circuit switching. *

☐ A

☐ B

☐ C

☒ D

 Will be reviewed

17

Consider the following diagram, what is the destination MAC address in the frame transmitted from node A if node A is sending to node B?

- A) 74-29-9C-E8-FF-55
- B) E6-E9-00-17-BB-4B
- C) 1A-23-F9-CD-06-9B
- D) 49-BD-D2-C7-56-2A *

☐ A

☒ B

☐ C

☐ D

 Will be reviewed

18

Which of the following is present in both HTTP request line and status line?

- A) HTTP version number
- B) Request method
- C) Status code
- D) URL *

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

19

Which of the following HTTP protocols does the given figure represent to fetch a web page with 2 reference objects?

- A) Non-persistent HTTP.
- B) Persistent HTTP.
- C) Non-persistent HTTP with parallel connection.
- D) Persistent HTTP with pipelining. *

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

20

Considering the operation of a learning switch and its forwarding table in the following figure. Which interface will the switch forward a frame transmitted from node D to node A?

- A) 1
- B) 3
- C) 4
- D) 1, 2, 4 *

☒ A

☐ B

☐ C

☐ D

 Will be reviewed

21

Consider a 4-bit generator G with value **1001**, what is the CRC checksum R if data D has the value **10001100001**?

A) 011

B) 100

C) 110

D) 0110 *

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

22

Given a subnet with network prefix 192.168.1.0/24, how many hosts can be connected to this subnet?

A) 256

B) 255

C) 253

D) 254 *

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

23

Which of the following statements is true about TCP and not true about UDP?

- A) Provides timing service.
- B) Provides minimum throughput guarantee service.
- C) Provides security service.
- D) Provides reliable transport service. *

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

24

Which of the following statements is true about circuit switching?

- A) Resources are pre-allocated regardless of demand
- B) Packet transmission can use the full link bandwidth
- C) The path between a source and a destination is not fixed
- D) All of the options *

☒ A

☐ B

☐ C

☐ D

 Will be reviewed

25

Which of the following layers doesn't appear in the Internet Protocol Stack?

- A) Application layer
- B) Network layer
- C) Transport layer
- D) Session layer *

☐ A

☐ B

☐ C

☒ D

 Will be reviewed

26

Which of the following statements about DNS is TRUE?

- A) TLD servers store all the hostname to IP mappings of the Internet.
- B) Every Web server must have a canonical name.
- C) Local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.
- D) DNS server listens to TCP port 53. *

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

27

Which of the following protocols can be used by routers for error signaling?

- A) DHCP
- B) ICMP
- C) ARP
- D) NAT *

☐ A

☒ B

☐ C

☐ D

 Will be reviewed

28

A Web server stores a webpage that comprises a base HTML file and 2 images referenced by the base HTML file. The HTML file is 100 bytes and each image is 200 bytes. A client is connected to the Web server through a direct link of 1 Mbps. Propagation delay between the Web server and the client is 50 milliseconds.

The client downloads the webpage using persistent HTTP but without pipelining (i.e. the next HTTP request is sent after the response for the previous HTTP request is received). Assume HTTP header, TCP header and ACK packets are of negligible size, transmission channel is perfectly reliable, time to establish and close TCP connection can be ignored. How long (in milliseconds) does it take for the client to download the entire webpage from the Web server?

- A) 404
- B) 604
- C) 204
- D) 402.4 *

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

29

A source and a destination are separated by 3×10^5 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long.

Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?

A) 200

B) 500

C) 100

D) 400 *

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

30

A Selective Repeat sender just receives an ACK packet with ACK number 7. This ACK number falls within sender window which has the window size 3. Every data packet embeds a k-bit sequence number field (k is a constant unknown to you). Which of the following definitely CANNOT be the sequence number of the next packet transmitted by the sender?

A) 0

- B) 2
- C) 3
- D) 6 *

☐ A

☐ B

☐ C

☒ D

 Will be reviewed

31

Consider a sender and a receiver communicating using Selective Repeat protocol. Every packet embeds a 4-bit sequence number field. Sender window size is 4. None of the packets shown in the following figure are corrupted packets. However, the third data packet and the second ACK packet are lost.

What is the sender window over the sequence number space at time t?

- A) 12, 13, 14, 15
- B) 13, 14, 0, 1
- C) 13, 14, 15, 0
- D) 13, 14, 15, 16 *

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

The router in the following figure is a NAT enabled router, what is the source IP address when the router forwards a datagram transmitted from the host with IP address 172.26.184.3 to a server with IP address 128.119.40.186.

- A) 172.26.184.3
- B) 128.119.40.186
- C) 172.26.184.1
- D) 137.132.228.5 *

☐ A

☐ B

☐ C

☒ D



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